N: Bhavar Kavan
S: 2503731070001
C: BE-CE-S4

Tutorial 5

B4.1 Land pollution

Q) What is land pollution? What are sources of land pollution

- Land pollution refers to degradation or contamination of earth's land surface due to human activities. It occurs when solid, liquid waste materials are dumped on land, reducing soil quality and making it unfit for agriculture and living organisms.

* Sources of land pollution

1) Industrial Waste
- Chemicals, Heavy metals, fly ash and toxic materials from factories.

2) Agricultural Activities : Excess use of fertilizers pesticides, ~~herbicide~~

3) Municipal Solid Waste
- Household garbage, plastics, food waste

4) Construction activities
- Cement, bricks, debris & demolition waste

5) Mining activities : overburden, toxic residues

6 Deforestation
- Removal of trees causing soil erosion

Q.2 How urbanization and construction causes land degradation?

- Urbanization refers to rapid growth of cities due to increasing population. It causes land degradation in following ways.

- Conversion of agricultural land into residential and industrial areas

- Increase in solid waste generation

- Reduction in green spaces and forests.

- Soil sealing due to roads, pavements and buildings

- Increased pressure on land resources.

* Construction Activities

* Construction causes degradation by :

- Removing fertile topsoil permanently
- Causing soil compaction due to heavy machinery
- Dumping construction debris in open land.
- Increasing dust production
- Altering natural drainage patterns

Q.3 What are causes of land pollution ?

- Major causes include :

1. Improper disposal of solid waste.
2. Excessive use of chemical fertilizers and pesticides
3. Industrial effluents and hazardous waste
4. Mining & extraction of minerals.
5. Deforestation and forest fires.

6. Rapid urban expansion

7. Overgrazing by cattle

8. Plastic waste accumulation

9. Landfill without proper lining system

10) Explain various effects of land pollution?

* Land pollution has various environmental and health impacts

1) Loss of soil fertility:
- Toxic chemicals destroy essential nutrient and microorganisms in soil

2) Water contamination
- Pollutants slip into groundwater, making drinking water unsafe.

3) Health problems
- Exposure to contaminated soil can cause:
  - Skin irritation
  - Respiratory problems
  - Poor air quality
  - Decomposing of matter, attracts harmful bacteria

4) Harm to wildlife
- Animals may consume plastic or toxic materials leading to their death

5) Soil erosion and Desertification
- Loss of vegetation leads to erosion and barren land

6) Climate Change
- Landfills produce methane or greenhouse gas.

7) Economic loss
- Reduces agricultural productivity affects farmers and income.

Q.5 Short note on control and prevention of land pollution.

- Land pollution can be controlled through.

1. Proper waste segregation
- Seperate biodegradable and non-biodegradable waste

2. Recycling and Reuse
- Promote 3 R principle:
  Reuse, Reduce, Recycle.

3) Composting
- Converting organic waste to compost

4) Organic farming
- Avoid harmful chemical fertilizers

5) Afforestation & Restoration
- Plant trees to protect soil

6) Scientific landfills
- Use proper lining system to prevent leakage

7) Strict Government
- Implement environmental regulations strictly

8) Public awareness program
- Educate people about environmental impact

9) Treatment of Industrial waste
- Treat waste water before disposal.

Waste